Informatica

Informatica® B2B Data Exchange
10.5.4

# Advanced Performance Tuning Guide

Informatica B2B Data Exchange Advanced Performance Tuning Guide
10.5.4
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# Preface

Use the *B2B Data Exchange Advanced Performance Tuning Guide* to learn how to configure the hardware and software that run B2B Data Exchange in PowerCenter workflows. Browse recommendations to improve throughput and scalability through benchmark tests that measure the performance of B2B Data Exchange and PowerCenter workflows.

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CHAPTER 1

# Performance Tuning

This chapter includes the following topics:


- Performance Tuning Overview, 7
- Hardware Configuration, 7
- Software Configuration, 9


## Performance Tuning Overview

The goal of performance tuning is to optimize throughput, which is the number of events that B2B Data Exchange processes every second.

Performance tuning involves adjustments to the hardware and software configuration in your environment, which assist you to accommodate the different throughput requirements in the organization. The hardware and software adjustments also help to improve the production environment scalability, which is the ability to manage the environment resources in a way that maintains the desired performance when the data volume increases.

This chapter contains recommendations for adjustments to your hardware and software configurations. The recommendations are based on benchmark tests that measure the performance and scalability of B2B Data Exchange and PowerCenter workflows on different databases, operating systems, and software configurations.

## Hardware Configuration

You can tune performance and optimize throughput by setting up and configuring your hardware to maximize throughput. Throughput is the number of events that B2B Data Exchange and PowerCenter process every second.

You can modify various hardware aspects to accommodate the required throughput. For example, you can install additional CPU cores, replace slower hard disks with faster and more reliable hard disks, or cluster multiple computers to handle the different product installations instead of a single computer.

## CPU Cores

To improve the performance of the machines on which B2B Data Exchange runs, you can incorporate additional CPU cores based on the required throughput. You determine how many CPU cores to use based on the throughput requirements in your organization.

The recommendations in this section address throughput for B2B Data Exchange. For more information about the CPU core recommendations for PowerCenter and Data Transformation, see the *PowerCenter Performance Tuning Guide* and the *Data Transformation Performance Benchmark Guide*.

You can add CPU cores to a single machine, or you can use multiple machines in a cluster configuration. If you add CPU cores to a single machine, each CPU core adds full processing power to the machine overall processing power.

The following table describes the recommended minimum CPU cores that B2B Data Exchange requires for different throughput:

| Events per Second | Minimum CPU Cores |
|---|---|
| 100 | 8 |
| 200 | 16 |
| 300 | 24 |
| 400 | 32 |
| Adding CPU cores to the machine may not accommodate processing of more than 400 events per second. | |

## Physical Storage and Memory

To improve the performance of the machines on which B2B Data Exchange and PowerCenter run, consider the following recommendations:

- Store the Data Exchange document store on a high-performance storage system, such as Network-Attached Storage (NAS), to increase the processing reliability and performance when PoweCenter processes large documents that B2B Data Exchange passes by reference. If you require higher throughput, do not use a Network File System (NFS).
- Store the database repository on a high-performance file system, such as Storage Area Network (SAN), to improve the data transfer speed and reliability.
- Store the B2B Data Exchange transaction log and the database repository on a high-performance hard drive, such as Solid-State Drive (SSD), to accommodate increased amounts of log entries.
- Run B2B Data Exchange on a machine with at least 8GB of RAM.

## Cluster Environment

To improve the performance of the machines on which B2B Data Exchange and PowerCenter run, consider the following recommendations:

- Install the B2B Data Exchange server on a computer cluster instead of on a single machine.
- Store the database repository on a separate machine instead of on the same machine on which B2B Data Exchange and PowerCenter run.

# Software Configuration

You can configure settings in B2B Data Exchange, PowerCenter, and the database repository software to tune performance and optimize throughput.

For example, you can pass documents from B2B Data Exchange to PowerCenter by reference, set up batch event creation requests that PowerCenter sends to B2B Data Exchange, or increase the maximum size of the redo log in the database repository.

The following sections describe the software modifications you can make to tune performance and increase throughput. The recommendations are based on performance benchmark tests in B2B Data Exchange and PowerCenter.

## B2B Data Exchange Settings and Utilities

To improve the performance of B2B Data Exchange, consider the following recommendations:

- Set the log to INFO mode instead of DEBUG mode. By default, the B2B Data Exchange log generates less log entries in INFO mode. You can set the log to DEBUG mode troubleshoot issues and set it back to INFO mode to improve performance.
- In the dxruntime.sh utility for the Linux, AIX, or Solaris operating system, or the dxruntime.bat utility for the Windows operating system, increase the Java heap size for the B2B Data Exchange server. For more information about the utility, see the *B2B Data Exchange Installation and Configuration Guide*.
- In the server and console copies of the dx-configuration.properties file, you can tune several properties to improve throughput. For more information about the configuration file, see the *B2B Data Exchange Installation and Configuration Guide*.
  - Increase the number of maximum statements in the dx.jdbc.maxPoolSize property to 200.
  - To improve file send performance set the dx.mft.pool.maxSessionsPerEndpoint default setting to allow five files to be sent concurrently.
  - Tune the outbound threads control with the dx.queue.pool.outbound.threads.core property and the dx.queue.pool.outbound.threads.max property default setting to allow up to 20 messages to be processed concurrently.
  - Tune the inbound threads control with the dx.queue.pool.internal.threads.core property and the dx.queue.pool.internal.threads.max property default setting to allow up to 20 messages to be processed concurrently.
- When you define File Receive or Managed File Transfer endpoints for documents larger than 4KB, choose to pass documents by reference instead of by value.

## B2B Data Exchange Transformations in PowerCenter Workflows

To improve the performance of PowerCenter workflows that use B2B Data Exchange transformations, consider the following recommendations:

- In the **Config Object** settings of each PowerCenter workflow, increase the number of partitions that PowerCenter uses when it processes files to a minimum of 8 partitions for a throughput of 100 events per second. For example, define 24 partitions for a throughput of 300 events per second.
- In the Generate_File_Path transformation, select the **Do not use a temporary folder in the generated file path** check box to copy the file directly to the Data Exchange document store instead of in a temporary location. By default, PowerCenter copies each file first to a temporary location and then to the Data Exchange document store. If you select this check box, PoweCenter maintains only one copy of each file.

- The Add_Document_To_Event transformation may significantly impact the workflow processing time. If you include the transformation in the workflow, you can minimize the performance impact in the following ways:
  - Add a document to an event only when a processing error occurs.
  - For a parent event with multiple child events, aggregate the documents to a single file and attach it to the parent event.
  - If the average row size in the document is smaller than 4KB, pass the document by value. This prevents PowerCenter from writing too many documents to the Data Exchange Document Store.
  - If the average row size in the document is larger than 4KB, pass the document by reference with the Generate_File_Path transformation.
- On an SQL server with the Windows operating system:
  - Use the Profile_Parameter transformation once in the beginning of the workflow to retrieve all the parameters instead of multiple times in the workflow to retrieve individual parameters.
  - The Set function of the Event_Attributes transformation may drastically reduce performence. If you need to use the function, you can set all of the attributes once for the parent event.
  - The Initiate_Correlation and Complete_Correlation transformations may drastically reduce performance.
- To optimize API calls to B2B Data Exchange from PowerCenter, set the PowerCenter Integration Service environment variable **DX_JCT_MAX_ROWS_IN_BATCH** to 10.

The recommendations in this section are based on Data Exchange transformation benchmark test results. For more information, see " Data Exchange Transformation Benchmark Tests" on page 15.

## Database Repository

To improve the database repository performance, consider the following recommendations:

- Increase the number of concurrent sessions that the database repository can run.

  **Note:** Only the database administrator can perform this action. For more information, consult your database administrator.
- For Oracle servers, increase the size and the number of instances of the redo log. Each time the log file reaches the maximum size, the database creates a new log file. Increase the log file size to reduce the number of log files that the database creates during run time. Increasing the number of instances of the log file enables load balancing and increases the availability of the log files. The benchmark tests that are described in this document used three instances of the redo log with a file size of 2 GB for each instance .
- When the number of events in the event table exceeds three million, archive the events by using Data Archive. Otherwise, the B2B Data Exchange console can run slower or the scheduled jobs can hang. For more information about how to archive the events, see the *B2B Data Exchange Administrator Guide*.

## B2B Data Exchange JMS Broker

To improve the performance of the B2B Data Exchange JMS broker, consider the following recommendations:

- If you split messages with one PowerCenter workflow process the messages with another PowerCenter, send the messages between the PowerCenter workflows in batches of at least 20 messages instead of sending each message separately.
- Disable persistency when you send messages between PowerCenter workflows.
- In the activemq.xml file, increase the memory limit and the storage size to match the JVM size. To do this, enter a higher value in the **memoryUsage limit** and the **storeUsage limit** parameters. For example, if the

average message size is 10 KB, and your organization requires the queue to hold up to 1,000 messages, set the memory limit to at least 10 MB.

- Tune the values for the failover URL in the JNDI connection. For example, to change the timeout, maximum attempts, and initial reconnect relay, you might use the following settings:

  ```
  jms.closeTimeout=30000&timeout=30000&maxReconnectAttempts=5&initialReconnectDelay=1000
  ```

- If you expect to create hundreds of AMQ connections either from B2B Data Exchange or from PowerCenter, set the AMQ broker to use the NIO configuration. Locate the following parameter settings:

  ```
  <transportConnector name="openwire" uri="tcp://0.0.0.0:18616"...
  ```

  Change to the following parameter settings:

  ```
  <transportConnector name="openwire1" uri="nio://0.0.0.0:18616?
  useQueueForAccept=false"/>
  ```

## Configuration for Thousands of Endpoints

To improve the performance of the B2B Data Exchange endpoints when working with thousands of endpoints, configure the following settings:

- Improve the file input processing speed.
  - Use a fast file system. Ensure that the file input folders reside on computers with a fast file system.
  - Configure endpoints to multiple folders to avoid any individual folders that contain thousands of files.
- Adjust the file endpoint settings.
  - Tune the value for time interval to scan a folder for new files. Change the dx.endpoint.file.scan.interval system setting from the default value of **5** to the maximum number of seconds that the customer accepts, for example **120**.
  - Tune the value for the number of files expected each time that B2B Data Exchange scans a folder for new files. If more files are sent during peak periods than this value, the files wait for subsequent scans, which improves performance. Set the dx.endpoint.file.scan.max parameter to the expected number, for example **25**, instead of the default value of **250**.
  - Tune the value for the file backlog to a lower number. The backlog is the set of files that DX buffers during processing. Set the dx.endpoint.file.backlog.max parameter to a lower value, for example **250**, instead of the default value of **5000**. To change the expiration time for the backlog, set the endpoint.file.backlog.timeout parameter, whose default value is **10**.
- Schedule the MFT endpoints that act as inbound client to collect files at an appropriate frequency to your communication needs. A schedule that is too short creates many requests to the remote server that fetch few, small files. A schedule that is too long might create large processing peaks.

# Informatica Managed FIle Transfer Configuration Tuning

To improve performance, the following table describes the B2B Data Exchange property settings to be tuned:

| Properties | Settings |
|---|---|
| dx.infamft.httpclient.maxConnPerRoute | Define the maximum number of HTTP requests to Informatica Managed File Transfer for the same project.<br>For example, if Informatica Managed File Transfer is concurrently running five jobs from B2B Data Exchange for the same project, then the setting should be greater than five. |
| dx.infamft.httpclient.maxconnectionTotal | Define the maximum number of total HTTP requests to Informatica Managed File Transfer.<br>The value should equal or exceed the total number of concurrent jobs launched from B2B Data Exchange that will run in Informatica Managed File Transfer. |
| dx.queue.pool.outbound.threads.max | The setting should have a higher value than the value for the property `dx.infamft.httpclient.maxConnTotal`. |
| dx.queue.pool.inbound.threads.max | Maximum number of Receive Jobs to trigger at a time. |

The following connection and project timeout tuning is suggested:

- Increase project timeout in B2B Data Exchange system properties.
- Increase number of retry attempts for Informatica Managed File Transfer resource.
- Zlib compression is preferred for the SFTP resource, if the remote server supports this compression option.
- Increase the number of retry attempts for Informatica Managed File Transfer resources.
- Increase the client timeouts for remote servers and server timeouts for Informatica Managed File Transfer resources.

The Admin Server configuration in Tomcat for Informatica Managed File Transfer should be tuned to increase the maximum number of threads and the connection timeout for Informatica Managed File Transfer.

For Informatica Managed File Transfer, the following database tuning is suggested:

- Increase the maximum database connection to at least 200.
- Increase the maximum idle connection to the maximum to at least 30 seconds.
- Increase the maximum wait time to at least 10,000 msec.

CHAPTER 2

# Performance Benchmark Tests

This chapter includes the following topics:


- Benchmark Tests Overview, 13
- Transformation Performance Benchmark Test, 14
- Data Exchange Transformation Benchmark Tests, 15
- B2B Data Exchange JMS Broker Benchmark Test, 20
- Informatica Managed File Transfer Benchmark Test, 20


## Benchmark Tests Overview

A benchmark tests measures the performance and scalability of B2B Data Exchange and PowerCenter workflows that use Data Exchange transformations.

The test results help determine which hardware and software configuration adjustments you can make to tune the performance and optimize throughput in B2B Data Exchange and PowerCenter.

This chapter includes the following benchmark test types:

- Performance benchmark. Tests a single PowerCenter workflow that uses Data Transformation parsers and Data Exchange transformations under different CPU core combinations on single and multiple machines.
- Data Exchange transformation benchmark. Tests individual transformations with different document sizes under different conditions, such as passing by value or by reference, on different server vendors and operating systems.
- B2B Data Exchange JMS Broker benchmark. Tests the performance of the B2B Data Exchange JMS broker with different transaction volumes.

## Transformation Benchmark Test Machine Specifications

The transformation benchmark tests were performed on single machines and on all machines in a cluster configuration. The test configuration contained one Oracle database machine and three identical application machines with B2B Data Exchange, PowerCenter, and Data Transformation.

The following table lists the test machine specifications for all test machines:

| Category | Specification |
|---|---|
| Server Model | HP ProLiant BL460c G7 ServerOracle database |
| CPU Cores | 2 X 8 Cores Westmere-EX 9350 1.73GHz |
| RAM | 48GB |
| Local Disk | 2 X 300GB 10,000RPM SAS hard disk |
| Shared Storage | HP P2000 disk array with 49 X 146GB 15,000RPM SAS hard disks |
| Operating System | Red Hat Linux 5.6 64-bit |

# Transformation Performance Benchmark Test

The Transformation performance benchmark test determined the optimal hardware configuration and PowerCenter partitions to use when you want B2B Data Exchange to process a specific throughput of events per second with PowerCenter and Data Transformation.

The test was performed on a single machine with an 8 CPU core processing power, and in a cluster configuration with a combined 16 CPU core processing power. B2B Data Exchange sent documents with different file sizes to PowerCenter, which processed the documents by value with the X12 Parser transformation.

For the single machine, PowerCenter was configured to use 32 partitions. For the machine cluster, each PowerCenter installation was configured to use 16 partitions, with a total of 32 partitions.

The following table describes the performance benchmark test results:

| File Size | Number of Machines | Events per Second | PowerCenter Partitions |
|---|---|---|---|
| 2KB | 1 | 385 | 32 partitions |
| **2KB** | **2** | **500** | **2 x 16 partitions** |
| 6.5KB | 1 | 189 | 32 partitions |
| 6.5KB | 2 | 367 | 2 x 16 partitions |
| 16KB | 1 | 98 | 32 partitions |
| 16KB | 2 | 187 | 2 x 16 partitions |

The table illustrates that the machines reached maximum throughput when both machines in a cluster use 32 PowerCenter partitions to process a 2K file size document with no processing errors.

## PowerCenter Workflow Components

The PowerCenter workflow that the performance benchmark test used includes multiple Data Exchange transformations and Data Transformation parsers. The workflow is provided with this document.

The following table lists the order of the workflow components:

| Order | Component |
|---|---|
| 1 | JMS Input |
| 2 | DX_Event_Details |
| 3 | DT_Streamer |
| 4 | DX_Create_Event<br>**Note:** PowerCenter divided the file between the PowerCenter partitions and created an event for each partition. |
| 5 | DX_Add_Document_To_Event |
| 6 | DT_X12_Parser |
| 7 | DX_Add_Document_To_Event |
| 8 | DX_Set_Event_Attribute |
| 9 | DX_Event_Details |
| 10 | File Output |

# Data Exchange Transformation Benchmark Tests

The Data Exchange transformation benchmark tests help to determine the performance of individual transformations under different conditions, such as row size, number of documents, and server vendor.

The following sections include tests that were performed on transformations that can impact overall performance. Transformations that do not create substantial load on the workflow are not included.

For more information about recommendations for performance tuning of specific Data Exchange transformations and PowerCenter workflows, see "B2B Data Exchange Transformations in PowerCenter Workflows" on page 9.

## DX_Add_Document_To_Event Transformation

The following table lists the transformation benchmark test results for an Oracle database with the Linux operating system:

| Processing Type | Files | Rows | Events | Row Size | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|---|---|
| By Reference to Different events | 10 | 10000 | 100000 | 1KB | 3 | 53 | 429 |
| By Reference to Different events | 1 | 100000 | 100000 | 1KB | 2 | 30 | 667 |
| By Value to Different Events | 10 | 10000 | 100000 | 1KB | 1 | 48 | 926 |
| By Value to Different Events | 1 | 100000 | 100000 | 1KB | 0 | 37 | 2667 |
| By Value to Different Events | 10 | 10000 | 100000 | 10KB | 3 | 5 | 540 |
| By Value to Different Events | 1 | 100000 | 100000 | 10KB | 2 | 57 | 565 |
| By Value to Different Events | 10 | 2000 | 20000 | 100KB | 3 | 41 | 90 |
| By Value to Different Events | 1 | 20000 | 20000 | 100KB | 3 | 10 | 105 |
| By Value to Different Events | 10 | 300 | 3000 | 1MB | 9 | 50 | 5 |
| By Value to Different Events | 1 | 3000 | 3000 | 1MB | 7 | 47 | 6 |
| By Value to Same Event | 10 | 300 | 3000 | 1MB | 5 | 43 | 9 |
| By Value to Same Event | 1 | 3000 | 3000 | 1MB | 6 | 25 | 8 |

The following table lists the transformation benchmark test results for a Microsoft SQL Server database with the Windows operating system:

| Processing Type | Files | Rows | Events | Row Size | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|---|---|
| By Reference to Different events | 10 | 10000 | 100000 | 1KB | 4 | 55 | 339 |
| By Reference to Different events | 1 | 100000 | 100000 | 1KB | 2 | 50 | 588 |
| By Value to Different Events | 10 | 10000 | 100000 | 1KB | 0 | 49 | 2041 |
| By Value to Different Events | 1 | 100000 | 100000 | 1KB | 0 | 31 | 3226 |
| By Value to Different Events | 10 | 10000 | 100000 | 10KB | 1 | 38 | 1020 |
| By Value to Different Events | 1 | 100000 | 100000 | 10KB | 1 | 33 | 1075 |
| By Value to Different Events | 10 | 2000 | 20000 | 100KB | 2 | 24 | 138 |
| By Value to Different Events | 1 | 20000 | 20000 | 100KB | 2 | 32 | 131 |
| By Value to Different Events | 10 | 300 | 3000 | 1MB | 7 | 40 | 7 |
| By Value to Different Events | 1 | 3000 | 3000 | 1MB | 2 | 0 | 25 |

| Processing Type | Files | Rows | Events | Row Size | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|---|---|
| By Value to Same Event | 10 | 300 | 3000 | 1MB | 4 | 8 | 12 |
| By Value to Same Event | 1 | 3000 | 3000 | 1MB | 5 | 40 | 9 |

## DX_Create_Event Transformation

The following table lists the transformation benchmark test results for an Oracle database with the Linux operating system:

| Files | Rows | Events | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|
| 10 | 100000 | 1000000 | 5 | 20 | 3120 |
| 1 | 1000000 | 1000000 | 5 | 22 | 3105 |

The following table lists the transformation benchmark test results for a Microsoft SQL Server database with the Windows operating system:

| Files | Rows | Events | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|
| 10 | 100000 | 1000000 | 3 | 47 | 4405 |
| 1 | 1000000 | 1000000 | 3 | 48 | 4386 |

## DX_Event_Attribute Transformation

The following table lists the transformation benchmark test results for an Oracle database with the Linux operating system:

| Operation | Attributes | Files | Rows | Events | Row Size | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|---|---|---|
| Get Event Attribute from Different Events | 1 | 10 | 10000 | 100000 | 10KB | 0 | 45 | 2222 |
| Get Event Attribute from Different Events | 10 | 10 | 1000 | 100000 | 10KB | 0 | 18 | 5556 |
| Set Event Attribute to Same Event | 1 | 10 | 10000 | 100000 | 10KB | 0 | 50 | 2000 |
| Set Event Attribute to Same Event | 10 | 10 | 1000 | 100000 | 10KB | 0 | 7 | 14286 |

| Operation | Attributes | Files | Rows | Events | Row Size | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|---|---|---|
| Set Event Attribute to Different Events | 1 | 10 | 10000 | 100000 | 10KB | 0 | 54 | 1852 |
| Set Event Attribute to Different Events | 10 | 10 | 1000 | 100000 | 10KB | 0 | 28 | 3571 |

The following table lists the transformation benchmark test results for a Microsoft SQL Server database with the Windows operating system:

| Operation | Attributes | Files | Rows | Events | Row Size | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|---|---|---|
| Get Event Attribute from Different Events | 1 | 10 | 10000 | 100000 | 10KB | 0 | 11 | 9091 |
| Get Event Attribute from Different Events | 10 | 10 | 1000 | 100000 | 10KB | 0 | 9 | 11111 |
| Set Event Attribute to Same Event | 1 | 10 | 10000 | 100000 | 10KB | 0 | 48 | 2083 |
| Set Event Attribute to Same Event | 10 | 10 | 1000 | 100000 | 10KB | 0 | 23 | 4348 |
| Set Event Attribute to Different Events | 1 | 10 | 10000 | 100000 | 10KB | 3 | 19 | 502 |
| Set Event Attribute to Different Events | 10 | 10 | 1000 | 100000 | 10KB | 2 | 40 | 625 |

## DX_Event_Details Transformation

The following table lists the transformation benchmark test results for an Oracle database with the Linux operating system:

| Operation | Files | Rows | Events | Row Size | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|---|---|
| Get Event Details | 10 | 100000 | 1000000 | 10KB | 1 | 50 | 9091 |
| Set Event Details | 10 | 100000 | 1000000 | 10KB | 5 | 30 | 3030 |

The following table lists the transformation benchmark test results for a Microsoft SQL Server database with the Windows operating system:

| Operation | Files | Rows | Events | Row Size | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|---|---|
| Get Event Details | 10 | 100000 | 1000000 | 10KB | 2 | 29 | 6711 |
| Set Event Details | 10 | 100000 | 1000000 | 10KB | 4 | 7 | 4048 |

## DX_Generate_Temporary_File Transformation

The following table lists the transformation benchmark test results for an Oracle database with the Linux operating system:

| 'Do not use a temporary folder in the generated file path' check box | Files | Rows | Events | Row Size | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|---|---|
| Selected | 1 | 10000 | 100000 | 10KB | 2 | 7 | 787 |
| Cleared | 1 | 10000 | 100000 | 10KB | 3 | 22 | 493 |

## DX_Increment_Profile Transformation

The following table lists the transformation benchmark test results:

| Database | Operating System | Files | Rows | Events | Row Size | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|---|---|---|
| Oracle | Linux | 1 | 100000 | 100000 | 10KB | 1 | 10 | 1428 |
| Microsoft SQL Server | Windows | 1 | 100000 | 100000 | 10KB | 1 | 5 | 1538 |

## DX_Initiate_Correlation and DX_Complete_Correlation Transformations

The following table lists the transformation benchmark test results:

| Database | Operating System | Files | Rows | Events | Row Size | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|---|---|---|
| Oracle | Linux | 1 | 5000 | 5000 | 10KB | 0 | 12 | 416 |
| Microsoft SQL Server | Windows | 1 | 5000 | 5000 | 10KB | 3 | 14 | 26 |

## DX_Profile_Parameter Transformation

The following table lists the transformation benchmark test results:

| Database | Operating System | Files | Rows | Events | Row Size | Minutes | Seconds | Throughput |
|---|---|---|---|---|---|---|---|---|
| Oracle | Linux | 1 | 100000 | 100000 | 10KB | 1 | 22 | 1219 |
| Microsoft SQL Server | Windows | 1 | 100000 | 100000 | 10KB | 2 | 15 | 741 |

# B2B Data Exchange JMS Broker Benchmark Test

The B2B Data Exchange JMS Broker benchmark test helps to determine the optimal hardware configuration to use when you want B2B Data Exchange to process a specific throughput of events per second with the B2B Data Exchange JMS Broker.

The following table describes the benchmark test results:

| Number of Messages in the Transaction | Actual Throughput |
|---|---|
| 1 | ~650 |
| 5 | ~2200 |
| 10 | ~3000 |
| 20 | ~4500 |
| 100 | - Producer ~4900<br>- Consumer ~4200 |

# Informatica Managed File Transfer Benchmark Test

The Informatica Managed File Transfer benchmark test measured the performance and scalability of Managed File Transfer and B2B Data Exchange when using MFT tasks to get and put files. The test results help determine which hardware and software configuration adjustments you can make to tune the performance and optimize throughput in B2B Data Exchange and Informatica Managed File Transfer.

## Informatica Managed File Transfer Benchmark Test Machine and Configuration Specifications

The following table lists the test machine specifications for the Informatica Managed File Transfer and B2B Data Exchange test server:

| Category | Specification |
|---|---|
| Operating System, Virtual Machine | Windows Server 2012 R2 Standard |
| RAM | 32 GB |
| CPU Cores | 8 Core 2.40 GHz Processor |
| Kernel Version | 9.3.9600.18821 |

The following table lists the test machine specifications for the remote test server:

| Category | Specification |
|---|---|
| Operating System, Virtual Machine | Windows Server 2008 R2 Standard |
| RAM | 16 GB |
| CPU Cores | 8 Core 2.40 GHz Processor |
| Kernel Version | 9.1.7601.23915 |

The following table describes the B2B Data Exchange configuration:

| Category | Description |
|---|---|
| Number of partners | 20 partners with one account |
| PowerCenter batch workflow | One workflow |
| Profiles | 20 profiles |

The following table describes the Informatica Managed File Transfer configuration:

| Category | Description |
|---|---|
| Number of web users | Four web users |
| Resources | Eight resources were tested with the following details:<br>- One FTP resource<br>- One FTPS resource using password<br>- One FTPS resource using certificate<br>- Three SFTP resources using passwords<br>- Three SFTP resources using certificates<br>- Two SFTP resources using public/private key |
| Endpoints | Twenty Informatica MFT Receive and Send endpoints were tested with the following details:<br>- Two FTP Remote Receive and Send endpoints<br>- Two FTPS Remote Receive and Send endpoints using password<br>- Two FTPS Remote Receive and Send endpoints using certificate<br>- Six SFTP Remote Receive and Send endpoints using passwords<br>- Four SFTP Remote Receive and Send endpoints public/private key<br>- Four Hosted Receive and Send endpoints<br>All the endpoints use the same workflow. |

# Informatica Managed File Transfer Sizing Benchmark

The Informatica Managed File Transfer sizing benchmark test helps to assess the optimal setup when you use Informatica Managed File Transfer with B2B Data Exchange.

The following table shows the configurations for the sizing tests:

| Configuration | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Servers | 1 | 1 | 2 | 2 | 2 | 4 | 4 | 8 | 16 |
| CPU cores | 4 | 8 | 8 | 16 | 16 | 8 | 16 | 16 | 16 |
| Memory (GB) | 8 | 16 | 16 | 32 | 64 | 16 | 64 | 64 | 32 |

The following table shows the throughput rate in MBs for the sizing tests for both inbound and outbound files:

| | Average MBs / file | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Files/hour** | **0.01** | **0.1** | **1** | **5** | **10** | **25** | **50** | **100** | **500** | **1,000** |
| **10** | 0.1 | 1 | 10 | 50 | 100 | 250 | 500 | 1,000 | 5,000 | 10,000 |
| **100** | 1 | 10 | 100 | 500 | 1,000 | 2,500 | 5,000 | 10,000 | 50,000 | 100,000 |
| **500** | 5 | 50 | 500 | 2,500 | 5,000 | 12,500 | 25,000 | 50,000 | 250,000 | 500,000 |
| **1,000** | 10 | 100 | 1,000 | 5,000 | 10,000 | 25,000 | 50,000 | 100,000 | 500,000 | |
| **2,500** | 25 | 250 | 2,500 | 12,500 | 25,000 | 62,500 | 125,000 | 250,000 | 1,250,000 | |
| **5,000** | 50 | 500 | 5,000 | 25,000 | 50,000 | 125,000 | 250,000 | 500,000 | | |
| **10,000** | 100 | 1,000 | 10,000 | 50,000 | 100,000 | 250,000 | 500,000 | | | |
| **25,000** | 250 | 2,500 | 25,000 | 125,000 | 250,000 | 625,000 | | | | |
| **50,000** | 500 | 5,000 | 50,000 | 250,000 | 500,000 | 1,250,000 | | | | |
| **100,000** | 1,000 | 10,000 | 100,000 | 500,000 | 1,000,000 | | | | | |

For example, for the test where an average of 10 files/hour were transferred, with an average file size of 0.01 MB, the equivalent of approximately 0.1 MB of files transferred per hour.

For the test where an average of 25,000 files/hour were transferred, with an average file size of 500 MB, the equivalent of approximately 1.25 TB of files transferred per hour.

## Informatica Managed File Transfer Recieve and Send Benchmark

The Informatica Managed File Transfer Send and Receive benchmark test helps to determine the throughput of files for Informatica Managed File Transfer Receive with B2B Data Exchange.

The following table describes the File Receive action benchmark test results for Informatica Managed File Transfer:

| Type | 50 KB File (Receive) | 250 KB File (Receive, Decrypt) | 1 MB File (Receive) |
|---|---|---|---|
| Number of Peak Hours | 7 | 7 | 7 |
| Number of Normal Hours | 3 | 3 | 3 |
| Total Nmber of Files | 249600 | 321789 | 99996 |
| Files/Normal Hours | 19200 | 24753 | 7692 |
| Files/Peak Hour | 38400 | 49506 | 15384 |
| Missed Files | 0 | 0 | 0 |
| % CPU (Normal Hour) | 60 | 78 | 75 |
| % CPU (Peak Hour) | 60 | 78 | 80 |
| Network - Normal Hour | 1 MB | 6 MB | 6 MB |
| Network - Peak Hour | 1.25 MB | 6 MB | 6 MB |
| Disk I/O (Normal Hour) | 1.25 MB | 6 MB | 6 MB |
| Disk I/O (Peak Hour) | - | 40 MB | 32 MB |

The following table describes the File Send action benchmark test results for Informatica Managed File Transfer:

| Type | 50 KB File (Send) | 1 MB File (Send) | 1 MB File (Send) | 1 MB File (Send) | 1 MB File (Send, Encrypt) |
|---|---|---|---|---|---|
| NuMBer of Peak Hours | 7 | 7 | 7 | 7 | 7 |
| NuMBer of Normal Hours | 3 | 3 | 3 | 3 | 3 |
| Total NuMBer of Files | 59995 | 59995 | 249600 | 399997 | 299990 |
| Files/Normal Hours | 4615 | 4615 | 19200 | 30769 | 23000 |
| Files/Peak Hour | 9230 | 9230 | 38400 | 61538 | 46000 |
| Missed Files | 0 | 0 | 0 | 0 | 0 |
| % CPU (Normal Hour) | 30 | 30 | 60 | 70 | 80 |
| % CPU (Peak Hour) | 40 | 40 | 90 | 95 | 100 |
| Network - Normal Hour | 200 KB | 1 MB | 2 MB | 2 MB | 2.5 MB |
| Network - Peak Hour | 300 KB | 1 MB | 4 MB | 8 MB | 4 MB |
| Disk I/O (Normal Hour) | 6 MB | 50 MB | 40 MB | 52 MB | 200 MB |
| Disk I/O Peak Hour) | 12 MB | 60 MB | 80 MB | 70 MB | 300 MB |

## Informatica Managed File Transfer Longevity Benchmark

The Informatica Managed File Transfer benchmark test helps to assess action longevity when you use Informatica Managed File Transfer with B2B Data Exchange.

The following table describes the longevity benchmark test conditions:

| Description | Amount |
|---|---|
| Total Duration in Hours | 8 |
| Number of Normal Hours | 6 |
| Number of Peak Hours | 2 |

The following tables describes the longevity benchmark test hardware configuration for Informatica Managed File Transfer and B2B Data Exchange servers:

| Informatica Managed File Transfer and B2B Data Exchange Servers | Specifications |
|---|---|
| OS | Windows Server 2012 R2 Standard |
| RAM | 32 GB |

| Informatica Managed File Transfer and B2B Data Exchange Servers | Specifications |
|---|---|
| CPU | 8 Core 2.40GHz Processor |
| Machine Type | Virtual Machine |
| Kernal Version | 9.3.9600.18821 |

The following tables describes the longevity benchmark test hardware configuration for the remote:

| Remote Server | Specification |
|---|---|
| OS | Windows Server 2008 R2 Standard |
| RAM | 16 GB |
| CPU | 2.10GHz (4 Core) |
| Machine Type | Virtual Machine |
| Kernal Version | 9.1.7601.23915 |

The following table describes the longevity benchmark test results for Informatica Managed File Transfer. The test included 414 GB of files, with a test duration of 8 hours, repeated over 4 day.

| File Size | Receiving Files During Normal Hours | Receiving Files During Peak Hours | Sending Files During Normal Hours | Sending Files During Peak Hours |
|---|---|---|---|---|
| 50 KB | 1000 | 2000 | 1000 | 2000 |
| 1 MB | 500 | 1000 | 500 | 1000 |
| 10 MB | 500 | 1000 | 500 | 1000 |
| 100 MB | 50 | 100 | 50 | 100 |
| 500 MB | 10 | 20 | 10 | 20 |
| 1 GB | 4 | 7 | 4 | 7 |
| 5 GB | 3 | 6 | 3 | 6 |

The following table describes the longevity benchmark test total results:

| Description | Normal Hours | Peak Hours |
|---|---|---|
| Total Files/Hour | 2067 | 4133 |
| Total Size Inbound/Hour | 35 GB | 51 GB |
| Total Size outbound/Hour | 35 GB | 51 GB |

| Description | Normal Hours | Peak Hours |
|---|---|---|
| Total Processed Files | 12402 | 8266 |
| Total Size (GB) | 210 | 102 |

The following table describes the timeline of completed events:

| File Size | 7:00 AM | 8:00 AM | 9:00 AM | 10:00 AM | 11:00 AM | 12:00 PM | 1:00 PM | 2:00 PM | Subtotal |
|---|---|---|---|---|---|---|---|---|---|
| 50 KB | 1000 | 1000 | 2000 | 2000 | 1000 | 1000 | 1000 | 1000 | 10000 |
| 1 MB | 500 | 500 | 1000 | 1000 | 500 | 500 | 500 | 500 | 5000 |
| 10 MB | 500 | 500 | 1000 | 1000 | 500 | 500 | 500 | 500 | 5000 |
| 100 MB | 50 | 50 | 100 | 100 | 50 | 50 | 50 | 50 | 500 |
| 500 MB | 10 | 10 | 20 | 20 | 10 | 10 | 10 | 10 | 100 |
| 1 GB | 4 | 4 | 7 | 7 | 4 | 4 | 4 | 4 | 38 |
| 5 GB | 3 | 3 | 6 | 6 | 3 | 3 | 3 | 3 | 30 |
| Total | 2067 | 2067 | 4133 | 4133 | 2067 | 2067 | 2067 | 2067 | 20668 |

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